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Exercise 1

A. What is meant by L*oop-carried* dependency? Please provide an example with a short code snippet that demonstrates a loop-carried dependency that cannot be parallelized. (1)

B. Could you please explain what the aim of agglomeration is in the PCAM design methodology? (1)

C. Given the following C-based code:

```
int acc = 0;

for (i = 0; i < 100; i++)
   for (j = 1; j < 100; j += 2) {
      a[i][j] += a[i][j-1];
      acc += a[i][j];
   }

for (i = 0; i < 100; i++)
   for (j = 2; j < 100; j += 2) {
      c[i][j] += f(a[i][j]) + b[i][j-2];
   }
```

Where both matrices a, b and c store integers and already contain actual data (e.g., are not full of zeros).

How and under what conditions can the previous code be parallelized? Is there a way to write the code exploiting parallel OMP pragmas? (2)

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Exercise 2

A. Please describe the parallel scan pattern, including assumptions, PRAM complexity. In which other parallel patterns discussed during the lecture have been used? (2)

B. Please describe how the Split parallel pattern works and its performance considering a PRAM model. How is it implemented (pseudo code required)? (2)

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Exercise 3

A. **When** should you use the OpenMP `task` construct? Is it possible to replace this construct with the `section` construct? (If you say no, provide an example as justification) (1)

B. What is the tradeoff between using static and dynamic scheduling policies in a #pragma omp for worksharing construct? (1)

C. What is the parallelization issue within the following code fragment that aims to parallelize the pi computation using a Monte Carlo approach? (2)

```
float pi = 0;
#pragma omp parallel default(shared) {
 unsigned int seed = 0;
#pragma omp for firstprivate(seed) reduction(+:pi) {
for(int i = 0; i < 100; ++i) {
   const float x = (rand_r(&seed)) / (float) RAND_MAX;
   const float y = (rand_r(&seed)) / (float) RAND_MAX;
   if (x*x + y*y) pi += 1;
}}
pi = 4 * pi / 100;
```

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Exercise 4

A. Given the following signature of the MPI scatter function, why are `sendcount` and `recvcount` scalar? What if one or more processes need to compute leftovers? (1) **This question can be skipped if you got 1 or 3 points in the second challenge.**

```
int MPI_Scatter(const void *sendbuf, int sendcount, MPI_Datatype sendtype, void *recvbuf, int recvcount, MPI_Datatype recvtype, int root, MPI_Comm comm)
```

B. Why do you need `mpirun` to execute a valid MPI application? (1)

C. What are the two main strategies that I have to allocate a receiving buffer from point-to-point communication? (2) **This question can be skipped if you got 2 or 3 points in the second challenge.**

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